

Sarang Kailas Vehale

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Professional Summary

Engineer and researcher with an integrated B.Tech and M.Tech in cyber security, working across applied AI, scientific computing and Linux systems. Research internships at C-DAC, India's Ministry of Home Affairs (I4C) and QOSF covered verified numerical simulation of quantum systems, hybrid factorisation algorithms, language models for document analysis, and noise modelling in quantum protocols. Builds and maintains open-source tools in Python and Rust, and administers a bare-metal Linux server. Six peer-reviewed publications and preprints, four as first author.

Skills

- **Programming:** Python, Rust, C, C++, Bash, SQL, R, Java
- **AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face, LLM fine-tuning, retrieval-augmented generation, FAISS, LangChain, LangGraph, MCP tool integration, evaluation pipeline design, anomaly detection
- **Scientific Computing & Quantum:** NumPy, SciPy, Pandas, SageMath, mpmath, Qiskit, PennyLane, Lanczos methods, numerical linear algebra, semidefinite programming, exact diagonalisation, noise modelling
- **Systems & Tooling:** Linux (Arch, bare-metal administration), systemd, networking, Git, GitHub Actions CI/CD, Docker, FastAPI, REST APIs, AWS, Raspberry Pi (ARM)
- **Security:** applied and post-quantum cryptography, threat simulation (MITRE ATT&CK, CALDERA), incident response, OSINT, digital forensics, network security

Experience

C-DAC | Research Intern, Verification & Benchmarking Pune, India | August 2025 – March 2026

- Built CERTIFY-ED, a framework that computes each result for quantum many-body systems with independent engines (SageMath, NumPy, Lanczos) and accepts it only when they agree with each other and with high-precision references, to 1.6×10^{-14} .
- Extended it into symveig with symmetry-sector decomposition, and released both open source with an 81-test suite that reproduces bit-identically across laptop, HPC and CI. Two first-author preprints.

I4C, Ministry of Home Affairs | AI Engineering & Research Intern Delhi, India | December 2024 – March 2025

- Fine-tuned and retrieval-augmented multilingual language models to extract structured entities from forensic documents across several Indian languages.
- Built the evaluation pipelines that gated production rollout on accuracy and latency, with expert review of sensitive output.
- Built an OSINT platform for data collection, cleansing and entity correlation, and trained non-technical users on it.

C-DAC | Research Intern, Hybrid Quantum Algorithms Pune, India | May 2024 – July 2024

- Implemented and benchmarked hybrid quantum-classical factorisation (QAOA, Babai and Schnorr lattice methods) in Qiskit against classical baselines, and assessed the implications for RSA and post-quantum schemes. First and corresponding author on the resulting survey in SN Computer Science (Springer Nature).

QOSF | Research Intern, Quantum Cryptographic Protocols Remote | April 2024 – June 2024

- Modelled noise and decoherence in quantum teleportation protocols and improved fidelity 15% through entropy-based regularisation formulated as a semidefinite programme. Mentored by Dr. Carlos Kuhn, University of Canberra.

Bhoomi-Itech | Security Research Intern, Threat Simulation Delhi, India | May 2023 – July 2023

- Built a CALDERA and MITRE ATT&CK simulation environment from scratch on Linux and rewrote incident-response procedures on the findings.

Stanford Code in Place | Section Leader, CS106A (Python) Remote | 2025 and 2026

- Taught weekly sections and reviewed student code across two cohorts.

Projects

1. rdd: Memory-Safe Rust Reimplementation of GNU dd

- Rust rewrite of the Unix block-copy utility with BLAKE3 and SHA-256 verification in the copy path. MIT-licensed, GitHub Actions CI, github.com/sarang-kernel/rdd. Proposed upstream to GNU coreutils.

2. Multi-Agent Workflow System (LangGraph, MCP)

- Multi-agent system with persistent memory and tools exposed through MCP servers, running unattended on a schedule over live data sources and producing structured output.

3. personal-document-chatbot: Local RAG Pipeline

- Offline retrieval-augmented generation over a private document corpus using Ollama, LangChain and a FAISS vector store.

4. Host-Based Anomaly Detection over Linux Syscalls

- LSTM autoencoder over syscall streams from nginx, sshd and bash; cut false positives from 12% to 1.8% at 81% recall through systematic ablations on threshold selection.

5. Self-Hosted Server and Hardware Builds

- Administers a bare-metal Arch Linux server running Jellyfin, Audiobookshelf, AdGuard Home and Calibre-Web as native services. Built an e-reader and a custom mechanical keyboard, and works extensively with Raspberry Pi boards.

Publications

1. **Survey of Integer Factorization Using Lattice-Based Algorithms** S. Vehale, D. Kushwah, R. K. Neiwal, A. Saxena, SN Computer Science 7:655, Springer Nature, 2026. DOI: 10.1007/s42979-026-05231-x. First and corresponding author.
2. **CERTIFY-ED: A Multi-Layer Verification Framework for Exact Diagonalization of Quantum Many-Body Systems** S. Vehale, R. Goel, arXiv:2605.11787, 2026. First author; under review, SciPost Physics Codebases.
3. **symveig: Verified Eigenvalue Enclosures for Symmetry-Decomposed Hermitian Matrices** S. Vehale, R. Goel, arXiv:2606.16217, 2026. First author; submitted to Computer Physics Communications.
4. **Hierarchical Quantum-Accelerated Federated Learning for Scalable, Auditable, Cross-Enterprise AI Governance** S. Vehale, R. Vehale, International Journal of Scientific Research and Engineering Trends 11(5), 2025. DOI: 10.5281/zenodo.17441935. First author.
5. **Bits to Qubits and Shor** (book chapter) S. Vehale et al., in Quantum Computing and Machine Learning (QCML'25), CRC Press. First author; accepted, pre-publication.
6. **Some Results on Cumulative Residual Inaccuracy Measure of k-Record Values** R. Goel, V. Kumar, S. Vehale, T. C. Scott, Entropy 28(1):17, MDPI, 2026. DOI: 10.3390/e28010017.

Education

National Forensic Sciences University (NFSU) | B.Tech + M.Tech, Cyber Security (Integrated Dual Degree)

Delhi, India | 2021 – May 2026

Indian Institute of Technology Madras (IIT Madras) | B.S. Data Science (online degree programme)

Chennai, India | 2024 – May 2027 (expected)

Certifications

- Supervised Machine Learning (Stanford / DeepLearning.AI, Coursera); Machine Learning Foundations and Introduction to Cloud (AWS); SOC Foundations (Microsoft); OT Security Bootcamp (NFSU Centre of Excellence); Quantum Information Science and Engineering (Virginia Tech and C2QA); Quantum Computing (C-DAC Hyderabad and IIT Roorkee); Quantum Algorithms and QBronze, Qiskit (Womanium / WISER 2025).

Leadership & Activities

- Open source: proposed rdd upstream to GNU coreutils through the project mailing list; contributor through Hacktoberfest and GSSoC.
- Internet-governance fellowships: InSIG (India School on Internet Governance), AuIGF (Australia Internet Governance Forum), and the Youth IGF organising committee; contributed to DPDP Act discussions.
- All India Forensic Science Conference 2025: organiser and liaison officer for 200+ professionals. NCC Junior Under Officer: led a 540-cadet Army Attachment Camp (5 Sikh Light Infantry).

Languages: English, Hindi, Marathi, Sanskrit, Japanese • Interests: Linux systems, self-hosting, open source